

17. Light-year is a unit of which of the following?

- (a) Distance
- (b) Time
- (c) Pressure
- (d) Power
- (e) None of the above / More than one of the above

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Ans. (a)

Light year is the unit of distance. It is the distance, the light can travel in one year. The Astronomical unit is defined as an average distance between earth and sun.

18. What is the unit of pressure?

- (a) Newton / sq. meter
- (b) Newton-meter
- (c) Newton
- (d) Newton/meter
- (e) None of the above / More than one of the above

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Ans. (a)

The Pascal (Pa) is the unit of pressure or stress in the International System of Units (SI). It is named after the scientist Blaise Pascal. One Pascal is equivalent to one newton (N) of force applied over an area of one square meter (1m^2).

$$1\text{ Pa} = 1\text{ Newton/meter}^2 = 1\text{ kg} / \text{ms}^2 = 1\text{ kg m}^{-1}\text{s}^{-2}$$

19. The unit of pressure is :

- (a) kg/cm^2
- (b) kg/cm
- (c) kg/mm
- (d) kg/cm^3
- (e) None of the above/More than one of the above

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Ans. (e)

See the explanation of the above question.

20. Which one of the following quantities does not have unit?

- (a) Stress
- (b) Force
- (c) Strain
- (d) Pressure
- (e) None of the above/More than one of the above

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Ans. (c)

When the shape or size of a matter is changed by applying an external force, it is known as strain. Since, it is a ratio, hence it is dimensionless and without any unit. The unit of force is Newton (kg ms^{-2}) and the unit of stress and pressure is Pascal ($\text{kg m}^{-1}\text{s}^{-2}$).

21. Magnetic field is a

- (a) scalar quantity
- (b) vector quantity

- (c) dimensionless quantity
- (d) More than one of the above
- (e) None of the above

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Ans. (b)

A scalar quantity is defined as the physical quantity that has only magnitude on the otherhand, a vector quantity is defined as the physical quantity that has both magnitude as well as direction. Magnetic field at a particular point includes both magnitude and direction. Therefore magnetic field is a vector quantity.

Measuring Devices and Scales

1. On a rainy day

- (a) dry and wet bulb thermometers show identical readings
- (b) dry bulb thermometer shows lower reading than wet bulb thermometer
- (c) dry bulb thermometer shows higher reading than wet thermometer
- (d) More than one of the above
- (e) None of the above

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Ans. (c)

The Dry Bulb, Wet Bulb and Dew Point temperatures are important to determine the state of humid air. The Dry Bulb temperature, usually referred to as air temperature, is the air property that is most commonly used. When people refer to the temperature of the air, they are normally referring to its dry bulb temperature. The Dry Bulb Temperature refers basically to the ambient air temperature. It is called "Dry Bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air. On a rainy day dry bulb thermometer shows higher reading than wet thermometer.

2. A new Central Ministry of Skill Development and Entrepreneurship was created in

- (a) 2013
- (b) 2014
- (c) 2016
- (d) 2017
- (e) None of the above / More than one of the above

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Ans. (b)

The Ministry of Skill Development and Entrepreneurship was set up as full Fledged Ministry of Government of India on 9 November, 2014 to coordinate all skill Development efforts across the country.

3. Which of the following is not correctly matched?

- (a) Voltmeter - Potential difference
- (b) Ammeter - Electric current
- (c) Potentiometer- Electromotive force
- (d) Galvanometer - Electric resistance
- (e) None of the above / More than one of the above

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Ans. (d)

Galvanometer is used to measure a small electric current or its function. Electric resistance is measured by an ohmmeter. The other question pairs are correctly matched.

4. Which of the following is used to measure the intensity of earthquake?

- (a) Richter scale
- (b) Mercalli scale
- (c) Great Harbour wave
- (d) More than one of the above
- (e) None of the above

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Ans. (b)

Mercalli scale is used to measure the intensity of earthquake whereas Richter scale measures magnitude of earthquake.

5. Hot-wire ammeters are used to measure

- (a) only DC
- (b) only AC
- (c) neither AC nor DC
- (d) both AC and DC
- (e) None of the above / More than one of the above

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Ans. (d)

The hot wire ammeter is a device used to measure AC or DC current intensity based on the thermal expansion of a wire that is heated due to the flow of the electric current through it, as the heat produced is proportional to the square of DC or rms value of AC.

6. Which instrument is used for measuring electric potential?

- (a) Potentiometer
- (b) Galvanometer
- (c) Voltmeter
- (d) More than one of the above
- (e) None of the above

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Ans. (d)

- Voltmeter is an instrument used to measure the potential difference. It is used between two points in an electric circuit to measure the potential difference between those two points. It is used in parallel.
- Potentiometer is an instrument used for the measuring unknown voltage by comparing it with the known voltage.
- Galvanometer is an instrument used to detect current in the electrical circuit.

From the given option both Potentiometer and Voltmeter are used to measure electric potential. Hence option (d) is the correct answer.

7. The device to measure electric current is –

- (a) Voltmeter
- (b) Ammeter
- (c) Voltmeter
- (d) Potentiometer
- (e) None of the above/More than one of the above

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Ans. (b)

An ammeter is a measuring instrument used for measuring electric current in a circuit. Electric current is measured in Ampere (A).

8. The resistance of an ideal voltmeter is

- (a) negative
- (b) infinite
- (c) zero
- (d) one

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Ans. (b)

Voltmeter measures the voltage in a wire. It is placed in parallel with the location being measured. An ideal voltmeter has a resistance of infinite ohm (Ω) because it need to stop any current from moving through it and taking away from what it's measuring.

9. Which instrument is used to measure humidity?

- (a) Hydrometer
- (b) Hygrometer
- (c) Pyrometer
- (d) Lactometer
- (e) None of the above / More than one of the above

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Ans. (b)

A Hygrometer is an instrument used for measuring the moisture content in the atmosphere which is also called as humidity in the air. Spectrometer is an instrument used to measure properties of light over a specific portion of the electromagnetic spectrum. A Eudiometer is a laboratory device that measures the change in volume of a gas mixture following a physical or chemical change.